POAMS: LECTURE 4
THE UNIVERSAL CONSTANT *c*

Having introduced the philosophical setting of Normal Realism for what follows, in which the role of the ***observer*** *is key, we are committed to interpreting the universal constant c as simply as a 'conversion factor' for converting observational distances into observational times, as opposed to its classical interpretation as 'the speed of light'. In this lecture, I will examine the history behind this classical interpretation and show that, even from a pragmatic point of view, to interpret c as a speed is an unnecessary assumption to make, a theoretical over-elaboration of the observational facts. This lecture is based on some of the material in Chapter Two of 'Light Speed…'*

1. Normal Realism and light

To recap on some of the points made in the last lecture, Normal Realism seeks to remove the final relics of Descartes' Cartesian separation (*Cartesian dualism*) of the **observer** from physical reality. Not least of these persistent relics is the classical 'speed of light' interpretation of the dimensional constant *c* in Einstein's second axiom for Special Relativity, which separates the object observed from the observer of the object in the Cartesian way. Hence, the classical Cartesian separation has been replaced by what is called the 'Einstein separation', the logical fallacy of which is the same as in Descartes' separation of observational reality into objects as we perceive them and those same objects as perceived by 'God'. This leads to all sorts of logical problems. For example, how can the mass of a distant star or galaxy be said to be a property 'relative to the observer' when the observer can know nothing about it until the light from that object reaches them which, in the 'travelling' way of thinking, may be æons after it left the object?

Thus, despite there being no possible way known to empirical science in which the light by which we see things can be observed or detected 'travelling in *vacuo*', we nevertheless assume that our ideas of what a thing is 'in itself' and our ideas of 'how we perceive it' are fundamentally separated, by this metaphysical agency of light travelling. From a commonsense point of view, it may seem as though light is propagated through empty space with a constant speed, until we try to construct in our minds any intuitive idea as to just how that motion takes place.

In the new paradigm, the *light* that was historically regarded as an incidental mediator of information about an inexorably remote 'world beyond' becomes *that very information itself*, which is the fount of all our knowledge of space, time, matter and motion. In this new way of thinking, we cease trying to match our human perceptions and conceptions to the world as 'God' might be presumed to perceive it, and settle for solving problems arising from *logically conflicting interpretations* of ordinary and instrumental experience. This prompts us to revise our ideas in the various categories of observation and conception towards satisfying the need for what is commonly known as *understanding*.

It is informative to note that Einstein's justification for his own 'relativistic' interpretation of mas, length and time was that it was based squarely on the phenomenalist philosophy of Mach. However, Mach eventually rejected Einstein's Relativity, declaring in his view that it was a 'mere transitory inspiration in the history of science'. For Mach, Einstein's theory was not a consistent phenomenalism but an eclectic compromise between opposing ideas, which therefore made no logical or philosophical sense. At the heart of this inconsistency lies Einstein's absolutistic assumption of 'light travelling' invisibly in *vacuo*. The point of Mach's objection was that light 'travelling in *vacuo*' is impossible to observe, whereas observation is the very basis of relativity. The idea that 'light travels' also makes no logical sense, since it would represent an absolute motion with respect to the void itself.

Hence, the question arises as to why Einstein, along with the classical physicists, made this assumption about light. I shall give an answer to this question in what follows.

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2. <u>The Constant *c*</u>

In 1676, the Danish astronomer Olaus (or Ole, or Olaf) Römer (or Rømer, or Roemer) (1644-1710), whilst studying the orbit of Io, one of the principal moons of Jupiter, at different times of the year, noted that the time between successive eclipses of the satellite by its parent planet became shorter as Earth approach Jupiter, and longer as Earth moved further away. Since the actual period of the satellite's orbit must remain constant, Römer concluded that this extra time taken between eclipses represented the time taken for the sunlight reflected off Io to ‘travel’ the extra distance across the diameter of Earth’s orbit. Knowing the time taken over the distance, Römer was able to calculate what became known as the ‘speed of light in space’. It is certainly not surprising that this constant, being the ratio of distance and time, was interpreted as a ‘speed’. Unfortunately, the accepted value for the diameter of Earth’s orbit at that time was incorrect, so that Römer’s data produced too low a value for that ‘speed’. Nevertheless, historically, Römer’s observations have been heralded as the first successful determination of the constant ‘speed of light’ c.

The important point to realise here is that what Römer actually calculated was simply a constant ratio, c, between measures of observational distance and observational time. Logically, however, although c has the *dimensions* of speed, it need not necessarily have been interpreted as a speed as such. The idea that light ‘travels’ in any accepted sense was pure supposition. We never observe light leave an object and ‘travel’ towards us in order for us to see that object. Nor do we ever see light actually ‘travelling’ between one object and another in the pure vacuum of space.

Other astronomers, such as the Astronomer Royal James Bradley (1692-1762), in 1727 subsequently refined the value of the constant ratio c and this was further refined in the first terrestrial experiments performed by Armand Fizeau (1819-1896), which employed a light interferometer. The most highly-regarded successful terrestrial measurements of the value of c were those performed by Albert A. Michelson (1852-1931), starting in 1878 and continuing until his death. In particular, his 1926 experiments employed an apparatus consisting of a source of light, several different many-sided mirrors, each of which could be rotated using an electric motor, and a distant plane mirror, approximately 35 kilometres from the source. The exact distance was measured to a very high level of accuracy by the US Coastal and Geodetic Survey Team. The speed of rotation of a particular many-sided mirror was adjusted so that light from the source, reflected by the distant mirror, could be recorded by a detector adjacent to the source. Knowing the speed of rotation of the many-sided mirror and the number of faces, Michelson was able to calculate the ratio of the optical distance to the time taken for the light from the source to be detected. In this way, the value of the ratio c could be determined. Michelson and his team obtained the result of $c \cong 3 \times 10^{8}$ metres per second. (Incidentally, Michelson was the first American to be awarded the Noble prize for Physics, for other work.) Although many different experiments have been performed since 1926, this remains the accepted approximate value of the constant c. Since 1983, the accepted exact value of c is 299,792,458 metres per second.

It must be remembered that what Michelson actually found was the so-called 'speed' c measured relative to his apparatus here on Earth. According to that same 'speed of light' assumption, c should be different for differently moving frames of reference. However, Einstein's second postulate for Special Relativity states that c has to be the same for *all* inertial observers, independent of their relative speed. But this defies all commonsense. According to our everyday experience, if we are travelling in a car at 80 kilometres per hour and another car passes us at 100 kilometres per hour, then its speed relative to us is 20 kilometres per hour. Even if this simple relationship ceases at much higher relative speeds, it makes no sense for one particular speed to be the same for any observer, independent of their speed! Yet Einstein's second postulate tells us that if we were able to travel at, say, $0.9c$, a passing light signal would still have a 'speed' c relative to ourselves!

Since this is so obviously counter-intuitive, it is necessary to understand why Einstein postulated the universal constancy of the 'speed of light'. One of the main reasons comes from considerations involving electromagnetism.

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3. Electromagnetism and Maxwell's Equations

During the first half of the nineteenth century, the experiments of Hans Christian Ørsted (Oersted) (1777-1851) showed that electric currents produce magnetic fields. Jean-Baptiste Biot (1774-1862) and Félix Savart (1791-1841) later formulated the *Biot-Savart law*, together with its associated equation, which showed just *how* magnetic fields are related to electric currents. Michael Faraday (1791-1867) discovered that changing a magnetic field produces an alternating electric current. The work of these scientists demonstrated that the phenomena of electricity and magnetism are intimately connected, and so emerged the study of 'electromagnetism'.

In 1856, the ratio of the unit of 'charge' in electrostatic units to that in electrodynamic units was determined by Wilhelm Weber (1804-1891) and

Rudolf Kohlrausch (1809-1858). They found that this ratio was remarkably close to $c/\sqrt{2}$, where c had the known value and dimensions of what was then known as the 'speed of light'. However, the intention of the Weber-Kohlrausch experiment had certainly not been aimed at determining the 'speed of light', although this ratio was later interpreted as such. Indeed, originally, the constant $c/\sqrt{2}$ was simply called 'Weber's constant'.

It was the mathematician James Clerk Maxwell (1831-1879) who finally provided the explicit link between the new electromagnetism and optics. This was in a series of papers, published between 1862 and 1865, which culminated in *Maxwell's equations*. These equations (for 'free space') may be written in the form

$$\nabla.\mathbf{B} = 0, \text{ (a) } \nabla.\mathbf{E} = 4\pi\rho, \text{ (b)}$$

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t}, \text{ (c) } \nabla \times \mathbf{B} = \frac{1}{c}\left(\frac{\partial \mathbf{E}}{\partial t} + 4\pi\mathbf{j}\right) \text{ (d)}$$

In these equations, **E** is the *electric field intensity*, **B** is the *magnetic induction*, and **j** is the *current density*. All are vector functions of position and time. The scalar variable ρ is also a function of position and time and signifies the *charge density*. The important fact to note here is that the constant c in these equations is a multiple of the ratio determined by Weber and Kohlrausch, and was interpreted as 'speed of light'. Hence, according to classical physics, these equations hold only relative to one particular frame of reference. However, since the laws governing optics and electromagnetism cannot logically depend on the choice of (inertial) observer, Maxwell's equations dictate that the constant c must be the same for all (inertial) observers, and so indicate that there must be something seriously wrong with classical Physics.

It can be deduced from Maxwell's equations that, in the absence of charges and currents (so $\rho = 0$ and $\mathbf{j} = \mathbf{0}$), **E** and **B** satisfy the *three-dimensional wave equation*, some solutions of which have wave-like properties.

[$\rho = 0$ and $\mathbf{j} = \mathbf{0}$ in (a) – (d) with some manipulation involving vector identities gives:

$$\nabla^2\mathbf{B} = \frac{1}{c^2}\frac{\partial^2 \mathbf{B}}{\partial t^2}, \nabla^2\mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}]$$

This seems to imply that the existence of electromagnetic waves and so, if the constant *c* in equations (c) and (d) is interpreted as the 'speed of light', then this supports the theory that 'light' (meaning the whole of the electromagnetic spectrum) is propagated in *waves*, in much the same manner as sound. In this sense, Maxwell's work appears to support the *wave theory* of light, a version of which was initially proposed by Descartes in the seventeenth century. Of course, there is no such implication if *c* in Maxwell's equations is interpreted merely as a constant.

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4. <u>The luminiferous ether</u>

According to the wave theory of light, electromagnetic waves require a medium in which to propagate, in the same way that water waves require water and sound waves require air. However, it is obvious that light reaches Earth from the Sun and stars through the vacuum of space. Hence, a required hypothetical carrier of electromagnetic waves, known as the *luminiferous ether*, gained widespread acceptance.

It was then supposed that 'light travels in waves', regarded as 'stresses propagated through the ether'. It seemed natural to identify this 'ether' frame with absolute (stationary) space. The constant *c* in Maxwell's equations then became interpreted as 'the speed of light' relative to the 'ether frame'. A most urgent problem of the time was therefore the detection of Earth's motion 'through the ether'. The famous Michelson and Edward Morley (1838-1923) experiment, performed in 1887, set out to do just that. The experiment attempted to detect the ether by comparing the relative 'speeds of light' parallel and perpendicular to the 'ether wind' supposedly caused by Earth's motion through the ether. The results of this and other experiments, repeated over and over, were completely and consistently negative. The 'ether wind' remained undetected.

Reluctant though the classical physicists of the day were to accept this negative result, it appears that none of them thought of questioning the reliability of the experiment or its interpretation, such was the esteem with which Michelson was held. The simplest, most logical explanation for the null results is that either there is no absolute reference frame of the 'ether' or that, if the 'ether' does exist, its existence is undetectable. (Michelson was a firm believer in the existence of the ether and so always viewed his experiments as a failure. He never accepted Special Relativity.)

Nevertheless, because the existence of the 'ether' was so firmly accepted by the physicists of that time, other ether-based explanations for the negative Michelson-Morley results were suggested. One explanation, which received a lot of support, supposed that the null result was due to the *Lorentz-Fitzgerald contraction effect*. This was the suggestion that all lengths shorten in the direction of Earth's motion through the ether by a particular factor but that they remain unaffected in other directions. This theoretical contraction effect could never be directly detected, since any measuring device would also shrink by the same factor. The conclusion, therefore, was that the 'ether' certainly existed but was not detectable by any observation or experiment!

The logical conclusion from all this, then, especially from our Normal Realist point of view, is that any assumption as to the existence of an 'ether' is not only un-empirical but is also logically redundant. More importantly, if the constant c appearing in Maxwell's equations is interpreted simply as *dimensional constant of observation*, then there is no implication of 'light travelling in waves' and so no implication of an associated ether.

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5. The Normal Realist approach to the constant c

There are many other problems associated with the premise that 'light travels'. For example, as soon as this idea is accepted, there is the problem of the wave/particle dualist nature of light. Assuming that 'light travels', the quantised character of the observed spectral energies suggests that light 'travels as particles', whereas their period and resonant character suggest that light 'travels as waves'. This purely theoretical, dualistic behaviour of light leads to all sorts of conceptual problems.

Einstein's redundant additional interpretation of c as the 'speed of light' therefore, to say the least, appears rather odd, especially since one of the main achievements of Special Relativity was to make redundant the concept of the 'ether' as the hypothetical medium for the transport of 'light waves'. However, the essential ingredient for Special Relativity is that c is a constant relative to all inertial observers, not that it is a speed of any kind.

We would argue, then that in view of all the difficulties attending the 'speed' interpretation of *c*, it is much more conceptually logical and economical to interpret *c* as a *universal conversion factor* for converting optical distances measured, say, in units of metres of distance into units of light-seconds. Of course, in Astronomy, distances are commonly recorded by radar. As classically conceived, an observer measured the distance of an object by sending out a 'light signal' which is reflected off the object and received back by the observer. This distance is then determined simply by halving the time-difference between emission and reception. It would be impractical and illogical to attempt, in this way, to measure vast distances in kilometres by means of that signal, when it is the signal itself on which we depend for providing the information about the distance. For instance, imagine how absurd it would be to measure the 'speed of light' across the distance of a light-year!

We recognise that light does appear to 'travel' in the accepted sense, since a common opinion is that it can be 'observed doing so'. For example, in dusty atmospheres, interactions with dust-particles form 'rays' of illumination which seem to suggest that this illumination is 'travelling'. Some critics of our 'non-travelling light' approach have pointed to the fact that, as they see it, the light from a galactic supernova can be 'seen' progressing through cosmic dust-clouds, and so on. However, in the ultimate analysis, those 'rays' are no more than *phenomena* consisting of discrete atomic illuminations, giving the *impression* of motion, like the advertising signs that are lit by on-off sequences of flashing light-bulbs. Where there is no dust or anything else present (that is, truly a vacuum) no such 'beams' can be detected. Hence, to reiterate, light 'travelling' i*n vacuo*' is impossible to observe, both logically and empirically. As Bondi states in his book, 'Assumption and Myth in Physical Theory', Cambridge University Press, 1965, page 28:

'*Any attempt to measure the velocity of light is …not an attempt at measuring the velocity of light but an attempt at ascertaining the length of the standard metre in Paris in terms of time-units.*'

Aside from any practical, physical or philosophical objections, there are many other logical and mathematical reasons against interpreting the constant *c* as a 'speed'. Some of these reasons will be explored in later lectures. One argument against the 'speed' interpretation of *c*, involving an electrical circuit, is given in section 2.7 of 'Light-Speed…':

‘Light-Speed, Gravitation and Quantum Instantaneity’, Anthony D. Osborne and N. Vivian Pope, Phi Philosophical Enterprises (2007), p.33-37.

To summarise, in view of all these considerations, our Normal Realist replacement for Einstein’s postulate concerning *c* is:

Observational distance and time have a constant ratio of units, c, for all observers sharing that same conventional choice of units.

But why, it may be asked, does the constant ratio *c* have such a unique value of 2.99792458×10^8 metres to the second? The reason is simply the circumstantial choice of units for distance and time in our history of measurement. In hindsight, we could have chosen the standard unit of distance to be the light-second, in which case, *c* would have been unity and never appeared in Maxwell’s equations to raise the plethora of problems that it has.

The universal constant ratio *c* then is simply a ‘conversion factor’ for converting our observationally projected measures in conventional metres into conventional measures in seconds, in the way that one might convert measures in miles to measures in kilometres. However, the difference is that it is by no means easy to see how units of one measure, such as length, can be converted into units of a different measure, such as time. But this is only because we are long habituated to thinking of length and time as different and independent measures.

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6. Light as pure information

We maintain that our pure, theoretically unembroidered ‘universal conversion factor’ interpretation of the constant *c* unblocks the route to a natural acceptance of the relativism of Mach. But if the traditional mechanistic ‘streaming substance’ notion concerning light is relinquished, then how is light to be conceived according to Normal Realism? The answer is to think of light as *pure information.* This includes phenomena which are detected and revealed by the use of instruments of any kind, at all levels of the spectral range, from below infrared to beyond ultraviolet and gamma. It is well-known in modern physics that light is 'atomised' in ultimately discrete and irreducible energy-events called *quanta*. This *quantisation* of light will be fully explained in a later lecture. These quanta occur in our eyes and other instruments, like the pixels on a video screen, out of informational patterns and sequences of which the viewer projects a video scenario. Those patterns and sequences of quantum 'blips' are the ultimate *information* from which, as observers, we project the distances of objects from ourselves and from one another in the three-dimensional form we perceive as 'space'; and from the way those distances change, as measured by our clocks and other processes, we extrapolate the passage of 'time'. The masses of those objects can then be measured in the way in which, over time, they are seen to behave in relation to one another, in the same way that the masses of the Sun and planets are determined. Every physical property and quality that defines a physical object can ultimately be measured by these same optical methods, through the use of instrumentation. In the same way that other observers and objects are informational projections from our received light-information, so we are informational projections from their received information, all in the matrix of to-ing and fro-ing we know at first hand as 'objective space'. This community of perspectives disposes of the Subjective Idealism of the early phenomenalists, as discussed in the last lecture.

So, in light, taking its full spectral range, is contained all the information there is about the world and our place in it. This includes light that we, as human observers, can never see, but can detect through the use of instrumentation in all those ways known to natural science. In summary, in this new Normal Realist approach, light is not something that is just incidental to matter; it is basic to matter's very existence. This new approach to Physics may be described as *digital* or *informational*. In this new approach, there can be none of those gaps, voids or spaces that we ordinarily think of as separating the ultimate informational bits out of which all space and time are extrapolated. All such gaps or voids are observational projections *out* of the information. They cannot also be *preconditions* for the information out of which they are projected.

The light-in-space of the standard model becomes the *space-in-light* in this informational model, which may be conceived as somewhat in the nature of a hologram. The word 'hologram' as used here, does not signify, as in the usual, technological language, a ‘plate which gives rise to 3D images’. It signifies more of a *philosophical* concept, in which the emphasis falls on ‘holo-’, meaning *whole* and in which the dimensions of the optical images are dependent upon the manner in which optical information is observed.

Having presented the Normal Realist alternative to Einstein’s second postulate of Special Relativity, I will present the Normal Realist analogue of Special Relativity, which concerns uniform motion, beginning in the next lecture.

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